MarketSage AI

Master Engineering Blueprint — **Part IV**

Candle Projection, Options Engine and Risk

Honest fan-chart ghost candles, the implied volatility surface, structure selection, Monte Carlo P and L, position sizing, risk limits and kill switches.

This is an extract of the MarketSage AI master plan, published separately for convenience. Chapter, section and equation numbering matches the full document, and cross-references point back into it. Read Chapter 3 (Reality Calibration) first — every design decision in this part follows from it.

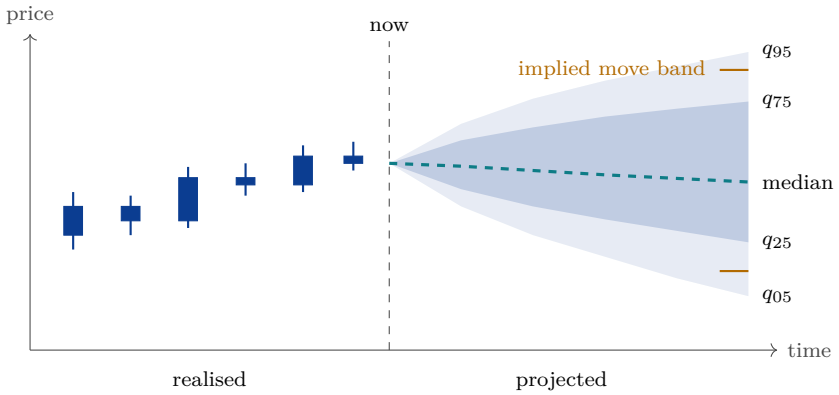
CHAPTER 10

Candle Projection: Honest Ghost Candles

10.1 The redesign

The original design draws a single predicted OHLC sequence. That is scientifically indefensible and, worse, psychologically dangerous: a crisp predicted candle invites the user to bet as if it were certain.

The redesign keeps the visual drama and adds truth: draw a **probability cone** made of quantile candles, where the body and wick widths encode uncertainty and the colour intensity encodes directional confidence.



10.2 How the cone is produced

1. **Volatility forecast** $\hat{\sigma}_h$ from M5 (HAR-RV plus boosted residual), conditioned on the event.
2. **Drift forecast** $\hat{\mu}_h$ from the calibrated direction ensemble converted to an expected return: $\hat{\mu}_h = (2p_{\text{up}} - 1) \cdot \mathbb{E}[|r_h|]$, damped by the meta-labeller's confidence.
3. **Direct quantile heads** from M2 give $q_\alpha(r_h)$ for $\alpha \in \{0.05, 0.25, 0.5, 0.75, 0.95\}$ at each horizon.
4. **Path sampling** for the visual: draw $N = 5000$ paths from a volatility-clustering process (a student-t innovation GARCH-style simulator, or a conditional generative model trained on real intraday paths) calibrated so the terminal distribution matches the quantile heads. Then compute the per-minute empirical quantiles of open, high, low and close.
5. **Rendering**: each projected minute is drawn as a semi-transparent candle whose body spans $[q_{25}, q_{75}]$ of the close and whose wick spans $[q_{05}, q_{95}]$, tinted by $P(\text{up})$.
6. **Overlay the implied move** from the option chain on the same axis. When the cone is narrower than the implied move band, options are expensive relative to our forecast — that is a sell-volatility setup, and it is immediately visible.

The killer visual

Two bands on one chart: *what we forecast* versus *what the market has priced*. The gap between them is the trade. No competing retail product shows this, and it makes the product's thesis legible in one glance.

10.3 Scoring the projection honestly

Metric	Definition	Target
Pinball loss	Quantile loss averaged over α	Must beat a random-walk baseline with GARCH volatility by at least 8% relative.
CRPS	Continuous ranked probability score	Lower than both random walk and the implied-volatility-derived distribution.
PIT histogram	Probability integral transform of realised outcomes	Should be uniform. Humps at the edges mean intervals are too narrow.
Empirical coverage	Fraction of outcomes inside $[q_{05}, q_{95}]$	Within $90\% \pm 3\%$.
Interval width efficiency	Mean width given coverage	Narrower is better only if coverage holds.
Directional deviation	$\lvert\text{realised} - \text{median}\rvert$	Reported, but never as the headline metric.

Reject the temptation

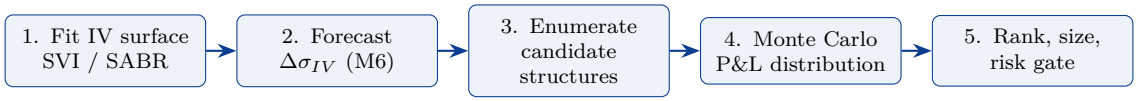
Do not display a single crisp predicted candle even if users ask for it. Instead display the median path as a thin dashed line inside the cone. The moment the product implies certainty, its users will size themselves into ruin and blame the product — correctly.

CHAPTER 11

Options Recommendation Engine

11.1 From a price forecast to a trade

A forecast distribution over the underlying is not a trade. The engine performs five steps.



11.2 Step 1 — The implied volatility surface

Raw per-strike implied volatilities from a live chain are noisy and occasionally arbitrageable due to stale quotes. Fit a smooth, arbitrage-free surface:

- Compute implied volatility from mid prices only (never last traded price — a stale last price generates a fake signal), using Black-76 on futures or Black–Scholes on spot with the appropriate forward.
- Filter: bid > 0, spread below a threshold, open interest above a threshold, and $|\delta| \in [0.05, 0.95]$.
- Fit a parametric slice per expiry (SVI is the standard choice), enforcing no calendar and no butterfly arbitrage.
- Store surface parameters, not just points. Slope, curvature and level become features; the fitted surface prices any strike, including illiquid ones.

11.3 Step 2 — Forecasting the volatility change

The most valuable model in the system

$\Delta\sigma_{IV}$ over the next 15–30 minutes is more forecastable than direction and it dominates short-dated option P&L. Typical patterns to learn:

- Implied volatility *expands* ahead of a scheduled event and *collapses* within minutes of the release — regardless of direction.
- After an unscheduled shock, implied volatility overshoots and mean-reverts with a half-life measured in tens of minutes.
- On expiry day, implied volatility decays mechanically and violently.
- Elevated implied volatility rank means buying premium has a strong headwind.

11.4 Step 3 — Candidate structures

Structure	Use when	Main risk	Notes

Long ATM/OTM call or put	Strong direction, forecast move well above implied, implied volatility rank low	Vega and theta	Simple, highest gamma, but the most exposed to volatility crush.
Vertical debit spread	Direction good, implied volatility elevated	Capped upside	**Default directional structure post-news.** The short leg neutralises much of the vega.
Long straddle / strangle	Big move expected, direction unclear, before a scheduled event	Volatility crush if the move disappoints	Only when forecast move exceeds implied move by a wide margin.
Short strangle with wings (iron condor)	Forecast move well below implied, calm regime	Tail risk	Defined risk only. Never naked short options.
Calendar spread	Term structure dislocated	Complex	Advanced; phase 3 of the options engine.
Ratio / broken-wing	Skew dislocation	Asymmetric	Advanced.
No trade	Forecast within noise of implied	—	Must be the most frequently selected option.

Strike selection is constrained, not free: only strikes with adequate open interest, adequate volume in the last five minutes, and a bid-ask spread below a fraction of the option price are eligible. A theoretically optimal strike that cannot be exited is a trap.

11.5 Step 4 — Monte Carlo P&L distribution

For each candidate structure:

1. Take the N sampled underlying paths from Chapter 10.
2. For each path and each future minute, evolve the implied volatility surface using the M6 forecast plus a stochastic component (correlated negatively with the underlying for equity indices — the leverage effect).
3. Reprice the structure along the path, including the passage of time.
4. Apply the exit rule (target, stop, time stop) to get a realised P&L per path.
5. Subtract full transaction costs on both legs, entry and exit, including modelled slippage as a function of the spread and size.

The output is a P&L distribution, from which we extract:

$$\mathbb{E}[\text{P\&L}], \quad P(\text{profit}), \quad \text{CVaR}_{5\%}, \quad \frac{\mathbb{E}[\text{P\&L}]}{\text{std}}, \quad \text{max drawdown along the path.}$$

Ranking objective

Rank by risk-adjusted expected value after costs, penalised by illiquidity:

$$\text{Score} = \frac{\mathbb{E}[\text{P\&L}] - \lambda\,\text{CVaR}_{5\%}}{\text{capital at risk}} \cdot \underbrace{\left(1 - \frac{\text{spread}}{2\,\text{mid}}\right)^2}_{\text{liquidity penalty}}$$

Emit a recommendation only if $\mathbb{E}[\text{P\&L}] > 0$ after costs *and* the score exceeds a minimum threshold *and* the meta-labeller approves.

11.6 Step 5 — Position sizing

Never size by intuition or by a fixed lot count. Use fractional Kelly with a hard cap:

$$f^* = \frac{p\,b - (1-p)}{b}, \qquad f_{\text{used}} = \min\left(0.25\,f^*,\ f_{\max}\right)$$

where p is the calibrated win probability, b the payoff ratio from the Monte Carlo, 0.25 is the fractional-Kelly discount (full Kelly is far too aggressive when p is estimated), and $f_{\max}$ is a hard per-trade cap (start at 1–2% of capital risked).

Additional size modifier	Effect
Posterior uncertainty of the impact prior	Size scales with $1/\sqrt{1+\text{var}}$; small sample sizes shrink the position.
Regime	Reduce size in high-volatility or low-liquidity regimes.
Time of day	Reduce size in the first 10 and last 15 minutes.
Consecutive losses	Reduce size after a losing streak (anti-martingale).
Correlation with open positions	Aggregate delta and vega across the book, not per trade.
Expiry day	Halve size; gamma risk is extreme.

11.7 Complete recommendation record

Every recommendation is an immutable, fully-explained record. This is both the product's differentiator and the audit trail.

Listing 11.1: Recommendation schema

```
{
  "reco_id": "uuid", "trace_id": "uuid", # links back to the news item
  "ts_emitted": "2026-08-26T09:20:41.318+05:30",
  "trigger": {"event_id": "...", "class": "geopolitical.military_action",
              "severity_cal": 8.2, "novelty": 0.91, "credibility": 0.88,
              "sources": ["src_a","src_b"], "seconds_since_event": 4.1},
  "forecast": {"horizon_min": 15,
               "p_up": 0.31, "p_up_ci": [0.24, 0.39],
               "quantiles": {"q05": -0.0142, "q25": -0.0071, "q50": -0.0033,
                             "q75": 0.0004, "q95": 0.0061},
               "sigma_forecast": 0.0089, "d_iv_forecast": +2.4,
               "implied_move_15m": 0.0052, "edge_ratio": 1.71},
  "structure": {"type": "bear_put_debit_spread",
                "legs": [{"side":"BUY","strike":19500,"opt":"PE","qty":1},
                         {"side":"SELL","strike":19300,"opt":"PE","qty":1}],
                "net_debit": 78.5, "max_profit": 121.5, "max_loss": 78.5,
                "breakeven": 19421.5},
  "greeks_net": {"delta": -0.21, "gamma": 0.0009, "vega": -1.8, "theta": -3.1},
  "mc": {"paths": 5000, "E_pnl": 21.4, "p_profit": 0.57,
         "cvar5": -71.2, "score": 0.184},
  "sizing": {"kelly_frac": 0.061, "used_frac": 0.015, "lots": 2,
```

```
            "capital_at_risk": 11775},
  "exits": {"target": 121.5, "stop": -39.0, "time_stop": "09:36:00",
           "vol_stop": "exit if IV drops > 3 pts"},
  "explanation": ["Military escalation, severity 8.2, first mention 4.1s ago",
                  "Historical prior for this class: median -0.31% over 15m (n=23)",
                  "Forecast |move| 1.71x implied move -> premium buying justified",
                  "IV rank 62 -> spread chosen over naked put to cut vega",
                  "Dealer gamma negative -> moves amplify"],
  "gates_passed": ["credibility>=0.8","two_sources","not_already_priced",
                   "liquidity_ok","daily_loss_limit_ok","regime_ok"],
  "model_versions": {"cls":"ev-cls-v7","dir":"ens-v12","vol":"har-gb-v4",
                     "iv":"ivd-v3","meta":"meta-v5"},
  "mode": "PAPER"
}
```

Why the explanation field is not decoration

Six months from now you will be trying to find out why the system lost money on a class of events. Without a stored, structured reason for every decision, that investigation is impossible and the system cannot improve. The explanation array is a first-class engineering requirement, not a UI nicety.

CHAPTER 12

Risk Management and Kill Switches

12.1 The risk gate sits between the brain and the money

No component may bypass the risk gate. It is deliberately simple, deliberately conservative, and deliberately hard to change (changes require a written justification recorded in the repository).

Limit	Starting value	Behaviour on breach
Per-trade capital at risk	1.5%	Reject or resize down.
Daily loss limit	3%	**Stop for the day.** Flatten everything. No overrides.
Weekly loss limit	6%	Stop for the week; mandatory review before restart.
Max concurrent positions	2	Reject new signals.
Net portfolio delta	bounded band	Reject trades that increase concentration.
Net portfolio vega	bounded band	Prevents accidental all-in volatility bets.
Max trades per day	6	Prevents overtrading in a noisy news day.
Max trades per event	1	Prevents pyramiding on one hypothesis.
Slippage guard	0.5% of premium	Cancel if fill price deviates.
Time-of-day block	first 5 min, last 10 min	No new entries.
Expiry-day multiplier	0.5x size	Gamma risk.
Model degraded flag	any	Signals become advisory only.
Data staleness	feed silent 3 s	Halt everything.

12.2 Kill switch hierarchy

Level	Trigger	Action
L1 Soft	Drift alert, calibration decay, unusual latency	Reduce size by half; notify.
L2 Pause	Daily loss limit, data quality failure, conflicting events	Stop new entries; manage existing positions only.
L3 Flatten	Feed loss, model server failure, clock drift, retraction of the triggering news	Close all positions at market immediately; stop.
L4 Lockout	Two L3 events in a week, or any unexplained loss	System refuses to run until a human clears it with a written root-cause note.

Physical kill switch

Provide a single command and a single dashboard button that flattens everything and disables the system, requiring no dependency on the very components that may have failed. Test it monthly. Also test what happens when the broker API returns an error during the flatten —

that is the scenario that actually costs money.

12.3 Scenario stress tests (run before each promotion)

1. Circuit breaker: the index hits a limit and trading halts mid-position.
2. Flash crash and instant recovery within 90 seconds.
3. Broker API returns success but the order was not placed.
4. Duplicate order acknowledgement (double fill).
5. Option chain goes stale but the spot feed stays live.
6. A false catastrophic headline followed by a retraction 4 minutes later.
7. Two contradictory high-severity events 20 seconds apart.
8. Expiry-day final 30 minutes with implied volatility collapsing.
9. Internet loss for 45 seconds with an open position.
10. Model server returns predictions from a stale model version.

For each: define the expected behaviour, implement it, and write an automated replay test. A system that has not been tested against these has not been tested.